

# ARI5118 - Quiz With Rationale

CNN Feature Visualisation and DeepDream

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**Quiz URL:** <https://forms.gle/tH9TxQb4cBkDk6zZ7>

## Question 1

**Correct answer: B.** Activation maximisation keeps the CNN weights fixed because the goal is to discover what input pattern already maximises a learned activation. The method probes the trained model rather than improving it.

**Plausible Pitfalls:** A is a common mistake because it describes training or fine-tuning, not visualisation. C is false because CNN weights can be updated during training. D sounds reasonable because both input and weights can be changed mathematically, but changing weights would alter the learned model rather than reveal what it already represents.

## Question 2

**Correct answer: B.** The main goal of activation maximisation is to generate or modify an input so that a selected neuron, channel, or class score activates strongly.

**Plausible Pitfalls:** A confuses activation maximisation with model training. C describes saliency or Grad-CAM, which explain existing images rather than synthesising preferred inputs. D describes pruning or compression, which removes parts of a model rather than visualising learned features.

## Question 3

**Correct answer: C.** The input image  $x$  is iteratively modified during activation maximisation. The network weights and biases remain fixed, while gradient ascent changes the input to increase the target activation.

**Plausible Pitfalls:** A and B are plausible because weights and biases are normally updated during CNN training. D is plausible because the learning rate or step size controls the update, but it is not the variable being optimised.

## Question 4

**Correct answer: A.** The gradient ascent update rule is:

$$x_{t+1} = x_t + \eta \frac{\partial a_i(x)}{\partial x} \quad (1)$$

is the basic gradient ascent update for increasing the activation of a chosen neuron or feature.

**Plausible Pitfalls:** B is a standard training equation because it updates weights using gradient descent on a loss. C reverses the derivative, which changes the meaning mathematically.

D updates the input but subtracts the gradient, making it gradient descent rather than gradient ascent.

### Question 5

**Correct answer: D.** Increasing the step size  $\eta$  makes each update larger. This can produce stronger visual changes, but it can also make the result unstable, noisy, or less interpretable.

**Plausible Pitfalls:** A is tempting because larger updates may look more dramatic, but they do not guarantee better interpretation. B incorrectly links step size to CNN depth. C confuses input optimisation with training, where model weights are updated.

### Question 6

**Correct answer: B.** Regularisation discourages unrealistic pixel-level artefacts and helps produce visualisations that are easier to interpret.

**Plausible Pitfalls:** A is plausible because classification confidence may increase, but regularisation does not force correct classification. C is false because gradient ascent is still required. D confuses feature visualisation with reconstruction of a training image.

### Question 7

**Correct answer: D.** CNNs usually learn a hierarchy of features. Early layers detect simple patterns such as edges and textures, while deeper layers combine these into more complex and object-like representations.

**Plausible Pitfalls:** A is false because feature hierarchy is not limited to medical CNNs. B reverses the usual order of abstraction. C sounds reasonable because deeper layers can be more task-specific, but layers do not all detect the same features.

### Question 8

**Correct answer: B.** Deeper layers are typically responsible for detecting semantic concepts, object parts, and class-related structures.

**Plausible Pitfalls:** A describes features more commonly associated with early or middle layers. C incorrectly suggests deep layers process raw pixels directly. D describes low-level features usually detected by early layers.

### Question 9

**Correct answer: C.** DeepDream produces repeated textures and object-like patterns because it amplifies activations in selected layers, reinforcing patterns already detected in the input image.

**Plausible Pitfalls:** A is a common misconception because DeepDream may look like learning, but the model is not retrained. B is plausible because the result looks artistic, but no handcrafted artistic filters are inserted. D is false because DeepDream does not retrieve or copy similar images from the training set.

### Question 10

**Correct answer: A.** Summing activations across a layer allows multiple features in that layer to influence the image update. This is why DeepDream can amplify several patterns at once.

**Plausible Pitfalls:** B sounds technical but confuses internal activations with class predictions. C is incorrect because DeepDream depends on gradients flowing back to the input. D is false because neurons are not learning new patterns; the trained model remains fixed.